PRIME MINISTER'S OFFICE,
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KOROGWE DISTRICT COUNCIL

MOMBO DIVISION JOINT EXAMINATION SYNDICATE
FORM FOUR BEST TEN STUDENTS EXAMINATION
032 CHEMISTRY 1

TIME: 3 HRS **DATE: TUESDAY 31ST MARCH, 2026**

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer all questions in sections A and B and **two (2)** questions from **section C**, question number **nine (9) is compulsory**
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section **C thirty (30) marks.**
4. Communication devices and any unauthorized materials are not allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).
6. All writing must be in **blue/black ink** except diagrams which must be drawn in pencil. Non-programmable calculators may be used.
7. The following constants may be used.
   Atomic masses: H=1, C= 12, K = 39, O = 16, Cl =35.5, Zn = 65.

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SECTION A (16MARKS)
Answer all questions in this section

1. For each of the items (i)–(x), choose the correct answer from the given alternatives and write its letter in the table provided below

   i. Calcium chloride crystals on a watch glass left overnight in the laboratory. Eventually a solution was found in the watch glass, this suggests that this compound is:
   A. Efflorescent B. Hygroscopic C. Deliquescent D. Solute E. Sublime

   ii. Kilango and Kimbi were arguing about the general formula of an alkyl group. Which among the following do you recommend to them?
   A. C_nH_{2n} B. C_nH_{2n-1} C. C_nH_{2n+2} D. C_nH_{2n+1} E. C_nH_{2n-2}

   iii. $10cm^3$ of 0.4M sodium hydroxide are added to $40cm^3$ of 0.2M hydrochloric acid. The resulting mixture will be:-
   A. Alkaline B. Neutral C. Acidic D. Distillation
   E. Separating funnel

   iv. At Taifa road junction, a truck carrying kerosene gets an accident, and unfortunately fire breaks out. People wanted to extinguish it using water but they were told not use water because water would spread the fire. This is because:
   A. Water is denser than kerosene B. Water is less dense than kerosene
   C. Water floats over the kerosene D. Water is neutral to fire
   E. Water is a universal solvent

   v. What is the IUPAC name of H_2S0_4
   A. Sulphuric (iv) acid B. Sulphuric (vi) acid C. Sulphuric acid

D. Hydrogen sulphate E. Hydrogen trisulphate

vi. Water can pass from atmosphere to land and back into the atmosphere. What term represents such a sequence?
A. Water purification B. Precipitation C. Evaporation
D. Water cycle E. Water treatment

vii. Which of the following sets of process use a gas that ignites with a pop sound when a lighted splint is passed through it?
A. Balloon filling, welding and diving
B. Hardening of oil, balloon filling and welding
C. Hardening of oil, balloon filling and diving
D. Fueling of rocket, diving and welding
E. Fueling car, diving and welding

viii. Why is wind said to be the promising source of energy for the future?
A. It does not produce harmful gases. B. It can easily be stored.
C. It is harnessed without chemical reaction.
D. It is a renewable source of energy. E. It is the cheapest source

ix. What is to be considered when choosing the best method to extract a particular metal from its ore?
A. The metal's ore impurities B. The metal shines C. The metal's economic value
D. Its availability in an area E. How it reacts with other materials.

x. Why is methane used as a source of fuel at homes and industries?
A. It burns in air to give out energy B. It burns in air to give out water vapour
C . It burns in air to absorb energy D. It burns in air to absorb water vapour
E. It burns in air to give out carbon dioxide

2. Match the effects of various chemicals on the living body in list A with their corresponding warning signs in list B by writing the letter of the correct response beside the item number in the answer booklet provided.

List A	List B
i. When poured on the wool, it completely damages the surface of the wool	A. Toxic
ii. When poured on the small fire, it bursts and turns a bigger fire	B. Radioactive materials
iii. When brought near the skin of someone, he/she starts to complain that his/her skin is itching	C. Corrosive
iv. When exposed on air, sudden blast accompanied with light occurs	D. Flammable
v. When poured into the food, the rat that eats it dies	E. Oxidant
vi. When exposed to you for a longtime, your organs will be damaged	F. Explosive
	G. Irritant
	H. Fragile

SECTIONB (54 Marks)
Answer all questions in this Section

3. Mr. Wachu made an investigation on the formation of reddish-brown coat on iron nail. He took a piece of iron nail and left it outside overnight for one day, on morning he saw brown coat on that piece of iron nail;
(a) Give the name of the brown coat formed
(b) Outline all necessary conditions for the formation of reddish-brown coat on iron nail
(c) Explain five methods of preventing the formation of reddish-brown coat although the all condition necessary for its formation are available

4. (a) We encounter many events/processes in our daily life that are good examples of chemical changes. Give three (3) examples of these changes.
   (b) List down three properties that clearly justify whether muddy water is a mixture or a compound.
   (c) Iron (III) chloride is an electrovalent compound which ionizes in water when electric current is allowed to flow through it.
      (i) How many moles of ions are formed when the compound is completely ionized in water?
      (ii) How many ions will be responsible for the passage of electric current in it?

5. (a) Form four students were taken to the chemistry laboratory for demonstrations of organic reactions between organic acids and alcohol. They found on the bench two reagents; butanoic acid and ethanol.
      (i) Write a chemical equation for this reaction
      (ii)Give the IUPAC name of the major product formed
      (iii) Which name is assigned to this type of reaction?
   (b) A hydrocarbon contains 80% carbon and 20% hydrogen by mass. Its' vapour density was measured and found to be 15. Calculate:
      (i) Its empirical formula
      (ii) Its molecular formula
      (iii) Which family of hydrocarbon does the molecular formula belong?

6. (a) A universal indicator has a pH values ranging between 1-14. Five solutions P, Q, R, S and T were subjected to it and their pH values read as follows:

Solution	P	Q	R	S	T
pH	01	05	07	09	14

   Which solution was:
      (i) Strong base
      (ii) Strong acid
      (iii) Weak acid
      (iv) Neutral solution
      (v) Weak base

   (b) A metal X (atomic number 11) burns in chlorine to produce a white solid chloride Y
      (i) By means of diagrams illustrate the arrangement of electrons in X before and after the reaction.
      (ii) Write a chemical equation for the reaction

7. (a) i) State the Le-Chatellier's principle
      ii) Any equilibrium reaction established must be associated with conditions. Name two of them.
   b) In the industrial preparation of Sulphur trioxide, equilibrium is established between Sulphur dioxide and oxygen gas as follows;
      $2SO_2(g) + O_2(g) \leftrightharpoons 2SO_3(g)$ $\Delta H = -84.9KJ/mol$
      i) How would you adjust temperature and pressure to maximize the proportion of the product at equilibrium?
      ii) What catalyst is used to speed up the rate of formation of Sulphur trioxide before attaining the equilibrium?

8. (a) Form four students Performed an experiment to determine hardness of water by using soap and the following:-
water sample.
Water sample A=water from Mkumbara
B=water from Chekelei
C=water from Buiko
D=water from Mazinde

Sample	VOLUME OF SOAP SOLUTION USED (cm^3)		
	Untreated H_2O	Boiled H_2O	H_2O passed through ion exchanger.
A	12	1.8	1.8
B	17	17	1.7
C	26	20	1.8
D	1.6	1.6	1.6

(a) Which water sample is hardest? Give reason for your answer.
(b) Which water sample behave as distilled water? Explain.
(c) Name a chemical substance that could be that cause of hardness in
(i) water sample A
(ii) water sample B
(d) which problems may face people who are using water from Mkumbara, Chekelei and Buiko?

SECTION A (30MARKS)
Answer ONLY two (2) questions in this section, question **nine (9) is compulsory**

9. A 6.3g of dibasic acid, **($H_2C_2O_4.XH_2O$)** was dissolved in water to make 1litre of a solution and 4.0g of NaOH dissolved to make 1 litre of a solution. The following results were obtained in the experiment when 25cm^3 of the basic solution was reacted with dibasic acid solution by using a POP indicator.

Burette readings (cm^3)	Pilot	1	2	4
Final burette reading (cm^3)	26.00	24.90	25.10	25.00
Initial burette reading (cm^3)	0.00	0.00	0.00	0.00
Volume of HCl (cm^3)				

(a) Complete the table above and Calculate the average volume of the dibasic acid used in the reaction.

(b) Write a balanced chemical equation for the reaction.

(c) Calculate the molarity NaOH

(d) Calculate the molarity **$H_2C_2O_4.XH_2O$**.

(e) Calculate the value of **X** in **$H_2C_2O_4.XH_2O$**

(f) Calculate the percentage of water of crystallization in **$H_2C_2O_4.XH_2O$**

10. A soil may be fertile but not productive. Discuss (five points)

11. Most of farmers in the Mombo divission were interested to know how the knowledge of chemistry is necessary in agricultural activities. Assume you are a chemist; educate these farmers on the application of chemistry in agriculture by giving five points.